Differential Expression Analysis

Statistical Concepts and Linear Models

UFHCI BCB-SR Bioinformatics Unit

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Roadmap

1. **The Problem**: Why is RNA-seq analysis challenging?
2. **Normalization**: Making samples comparable
3. **Design Matrices**: Encoding experimental structure
4. **Linear Models**: The foundation of DE testing
5. **The limma-voom Approach**: Handling count data
6. **Contrasts**: Asking specific questions
7. **Putting It All Together**: From counts to biology

The Challenge of RNA-seq Data

What Makes RNA-seq Different?

Count Data Properties:

- Discrete (not continuous)
- Right-skewed distributions
- Mean-variance relationship
- Zero-inflated
- Different library sizes

Implications:

- Can’t use normal distribution
- Can’t use simple t-tests
- Need specialized methods
- Must account for technical variation

→ Need: **Normalization** + **Appropriate statistical models**

The Library Size Problem

Sample A: 10M reads
Sample B: 20M reads

Gene X: - Sample A: 100 counts - Sample B: 200 counts

Question: Is Gene X differentially expressed?

Answer: Maybe not!

- Sample B has 2× more reads total
- Gene X has 2× more counts
- **Proportion is the same**: 100/10M = 200/20M

This is a **library size effect**, not biological!

→ Solution: **Normalization** to account for sequencing depth

The Composition Bias Problem

Scenario: Knockdown of a highly expressed gene

Before Normalization:

- Gene A (target): 50% of reads
- Gene B: 1000 counts
- Gene C: 500 counts

After Knockdown:

- Gene A: Now 10% of reads
- Gene B: 1200 counts (↑20%)
- Gene C: 600 counts (↑20%)

Problem: Genes B and C look "upregulated" but they're just taking up the "space" left by Gene A!

This is **composition bias** - a few genes can distort everything else.

→ Solution: **TMM normalization** (Trimmed Mean of M-values)

Normalization Methods

TMM Normalization

Goal: Find genes that are "stably expressed" and use them to calculate scaling factors

1. **Calculate M-values** for each gene: $M = \log_2(\text{Sample}_1/\text{Sample}_2)$
2. **Trim** the most extreme M-values (top/bottom 30% by default)
3. **Calculate weighted mean** of remaining M-values
4. **Derive normalization factor** from this trimmed mean
5. **Apply** to effective library sizes (not to the counts themselves!)

Key insight: Most genes should NOT be differentially expressed. TMM finds the "housekeeping" genes and uses them as reference.

What TMM Corrects

TMM corrects for:

- Library size differences
- Composition bias
- A few highly expressed genes
- RNA composition changes

TMM does NOT correct for:

- Batch effects
- GC content bias
- Gene length (already in counts)
- Biological variation

Assumptions:

- Most genes are NOT DE
- Symmetric changes (roughly equal up/down)

When TMM fails:

- > 50% genes are DE
- Very asymmetric changes
- Global shifts in expression

In practice, TMM works extremely well for most experiments!

Design Matrices

What is a Design Matrix?

Purpose: Encode your experimental design as numbers that statistical models can use

Example: Two groups (Control, Treated)

Sample	Group	Intercept	Treatment
S1	Control	1	0
S2	Control	1	0
S3	Treated	1	1
S4	Treated	1	1

R code: `model.matrix(~group)`

- Column 1 (Intercept): All 1's
- Column 2 (Treatment): 0 for control, 1 for treated

Reading a Design Matrix

The Formula:

For each gene:

$$Expression = \beta_0 + \beta_1 \times Treatment$$

- β_0 = Expression in control (intercept)
- β_1 = **Difference** between treated and control

For Control samples (Treatment = 0):

$$Expression = \beta_0$$

For Treated samples (Treatment = 1):

$$Expression = \beta_0 + \beta_1$$

Testing $\beta_1 = 0$?
→ "Is there a difference between treated and control?"
→ This is our **differential expression test**!

Two-Factor Design Matrix

Example: Genotype (WT, KO) × Treatment (Control, Drug)

```r
design <- model.matrix(~genotype + treatment)
```

Sample	Genotype	Treatment	Intercept	genotypeKO	treatmentDrug
S1	WT	Control	1	0	0
S2	WT	Drug	1	0	1
S3	KO	Control	1	1	0
S4	KO	Drug	1	1	1

The Model:

$$Expression = \beta_0 + \beta_1 \times KO + \beta_2 \times Drug$$

- β_0: Expression in WT + Control (reference)
- β_1: Effect of KO (averaged across treatments)
- β_2: Effect of Drug (averaged across genotypes)

Interaction Design Matrix

Question: Does the drug effect **depend on** genotype?

```r
design <- model.matrix(~genotype * treatment)
# OR: model.matrix(~genotype + treatment + genotype:treatment)
```

The Model:

$$Expression = \beta_0 + \beta_1 \times KO + \beta_2 \times Drug + \beta_3 \times (KO \times Drug)$$

- β_3 is the **interaction term**
- Tests: "Is the drug effect different in KO vs WT?"

If $\beta_3 \neq 0$: The drug works differently in KO!

- Example: Drug increases expression in WT but decreases in KO
- This is **divergent** response

Linear Models for RNA-seq

Why Linear Models?

Advantages:

1. **Flexible**: Can model any experimental design
2. **Multiple factors**: Genotype, treatment, batch, time, etc.
3. **Contrasts**: Ask any question post-hoc
4. **Borrowing strength**: Share information across genes
5. **Well-understood**: Decades of statistical theory

The Challenge:

- RNA-seq data are **counts**, not continuous measurements
- Counts have **mean-variance relationships**
- Can't use standard linear regression directly

→ Solution: **Transform** the data (voom) or **model counts directly** (edgeR/DESeq2)

Two Approaches to RNA-seq DE

Count-Based Models
(edgeR, DESeq2)

- Model counts directly
- Negative binomial distribution
- Dispersion estimation
- Exact statistical tests

Good for: - Simple comparisons - Small sample sizes - When normality questionable

Transform + Linear Model
(limma-voom)

- Transform counts to log-scale
- Use precision weights
- Standard linear models
- Empirical Bayes

Good for: - Complex designs - Contrasts and interactions - Leveraging limma ecosystem - Sample quality weighting

Good news: Both approaches give very similar results (~95% agreement)!

The limma-voom Approach

What is voom?

voom: "variance modeling at the observational level"

The Problem:

- High expression genes: stable, low variance
- Low expression genes: noisy, high variance
- This **heteroscedasticity** violates linear model assumptions

The voom Solution:

1. Transform counts to $\log_2$-CPM (counts per million)
2. Model the mean-variance relationship
3. Calculate **precision weights** for each observation
4. Use weights in linear model fitting

The voom Transformation

Step 1: Log Transform

$$y = \log_2(CPM + 0.5)$$

- CPM: Counts per million (normalized counts)
- +0.5: Avoid log(0)
- Makes data more “normal-like”

Step 2: Model Variance

$$\sqrt{\sigma^2} \sim \sqrt{y_{mean}}$$

- Plot √variance vs. mean
- Fit smooth curve (red line)
- This captures the relationship

Step 3: Calculate Weights

$$w = \frac{1}{\text{predicted variance}}$$

voomWithQualityWeights

Problem: Some samples might be outliers or poor quality

Solution: Calculate sample-level weights too!

1. **Gene-level weights** from mean-variance (standard voom)
2. **Sample-level weights** from overall quality

How it works:

- Samples with consistent, low-variance measurements: Weight ≈ 1.0
- Outlier samples with high variance: Weight < 1.0
- Exceptionally good samples: Weight > 1.0

Advantage: Automatically down-weights problem samples without removing them!

Empirical Bayes Moderation

The Problem: Individual gene variance estimates are unreliable

Without Moderation:

- Each gene: own variance
- Based on 3-6 samples
- Unstable estimates
- Noisy statistics

With Empirical Bayes:

- Pool information across ALL genes
- Estimate overall variance distribution
- "Shrink" individual estimates toward mean
- More stable statistics

Key Insight: Most genes have similar variance after accounting for mean expression. Borrow this information!

Result: Better p-values, more reproducible results

Contrasts: Asking Specific Questions

Why Contrasts?

Scenario: You have 4 groups (WT-Control, WT-Drug, KO-Control, KO-Drug)

Questions you might ask:

1. Does the drug work in WT?
   → `WT-Drug - WT-Control`
2. Does the drug work in KO?
   → `KO-Drug - KO-Control`
3. Does the drug work differently in KO vs WT?
   → `(KO-Drug - KO-Control) - (WT-Drug - WT-Control)`
4. Is there a genotype effect at baseline?
   → `KO-Control - WT-Control`

All of these can be tested from ONE model!

Contrast Matrix Anatomy

Example contrast: "Drug effect in WT"

```
makeContrasts(WT_Drug = WT_Drug - WT_Control, levels = design)
```

What this means:

	WT_Control	WT_Drug	KO_Control	KO_Drug	Contrast
Coefficient	0	0	0	0	
WT_Drug - WT_Control	-1	+1	0	0	Drug effect in WT

- Take mean of WT_Drug group: +1
- Subtract mean of WT_Control group: -1
- Ignore KO groups: 0

Any linear combination of group means can be tested!

Interaction as a Contrast

Question: "Does drug effect differ between genotypes?"

Step 1: Drug effect in WT
→ `WT_Drug - WT_Control`

Step 2: Drug effect in KO
→ `KO_Drug - KO_Control`

Step 3: Compare these effects
→ `(KO_Drug - KO_Control) - (WT_Drug - WT_Control)`

Expand the algebra:

$$(KO\text{-}Drug - KO\text{-}Control) - (WT\text{-}Drug - WT\text{-}Control)$$

$$= KO\text{-}Drug - KO\text{-}Control - WT\text{-}Drug + WT\text{-}Control$$

In contrast matrix:

Group	Coefficient
WT-Control	+1

Group	Coefficient
WT-Drug	-1
KO-Control	-1
KO-Drug	+1

Interpreting Contrast Results

logFC = +2.5

- Expression is $2^{2.5}$ = 5.7-fold higher
- In the "positive" group of contrast
- Biological: gene is upregulated

logFC = -1.8

- Expression is $2^{-1.8}$ = 0.29-fold
- In the "negative" group of contrast
- Biological: gene is downregulated

adj.P.Val < 0.05

- Statistically significant
- FDR-corrected (Benjamini-Hochberg)
- Accounts for multiple testing

adj.P.Val > 0.05

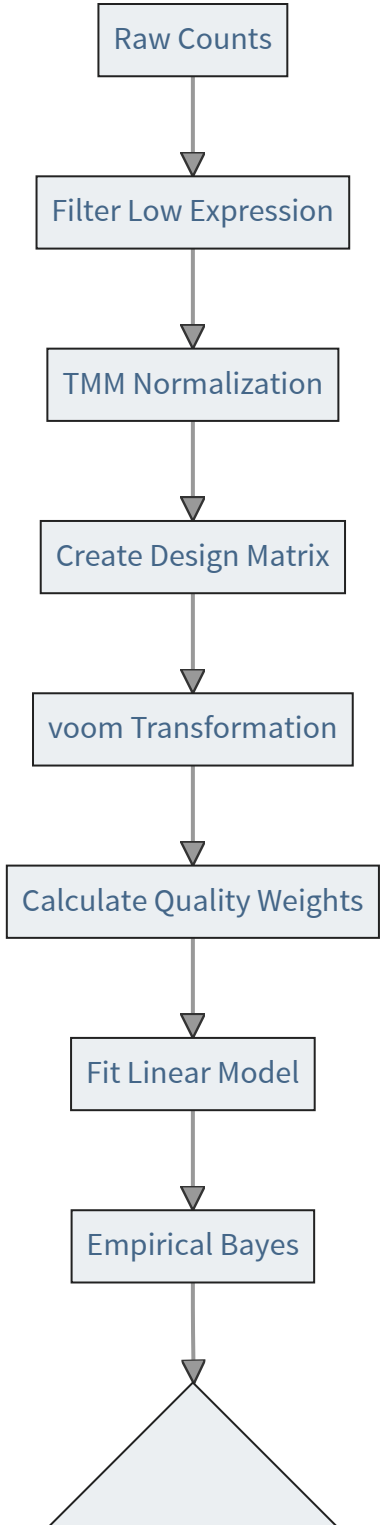
- Not statistically significant
- Could be no effect
- Or insufficient power

Always consider both:

- **Statistical significance** (adj.P.Val)
- **Biological significance** (logFC magnitude)

Putting It All Together

The Complete Workflow


Raw Counts
Filter Low Expression
TMM Normalization
Create Design Matrix
voom Transformation
Calculate Quality Weights
Fit Linear Model
Empirical Bayes

Key Decision Points

1. Filtering: How stringent?

- `filterByExpr()`: Sensible defaults
- More stringent: More power, fewer genes
- Less stringent: More genes, less power

2. Normalization: Which method?

- **TMM**: Default, works well for most
- **Upper quartile**: Very asymmetric DE
- **None**: If pre-normalized

3. Sample Weights: Use them?

- `voom()`: Standard, gene weights only
- `voomWithQualityWeights()`: +sample weights, more robust

Key Decision Points (cont.)

4. Model Design: Additive or Interaction?

- **Additive** (`~factor1 + factor2`): Assumes independent effects
- **Interaction** (`~factor1 * factor2`): Allows dependent effects
- Start simple, add complexity if needed

5. Contrasts: Which comparisons?

- Plan based on biological questions
- Can test multiple from same model
- Consider: main effects, interactions, specific groups

6. Thresholds: What is significant?

Common Pitfalls

Statistical Issues:

- ❌ Not filtering low-expression genes
- ❌ Forgetting normalization
- ❌ Wrong reference level in factors
- ❌ Testing >1 contrast without correction
- ❌ Ignoring batch effects

- ✅ Use filterByExpr()
- ✅ Always normalize (TMM)
- ✅ Set levels explicitly
- ✅ FDR correct across all tests
- ✅ Include batch in model

Interpretation Issues:

- ❌ P < 0.05 means "important"
- ❌ logFC = 0.5 is "not different"
- ❌ Non-significant = no effect
- ❌ More DE genes = better experiment
- ❌ Ignoring effect size

- ✅ Consider biology + statistics
- ✅ 0.5 = 1.4-fold change (real!)
- ✅ Could be low power
- ✅ Quality > quantity
- ✅ Report both logFC and p-value

Best Practices Summary

Questions to Ask Yourself

Before Running Analysis

1. **What is my biological question?**
    - Specific comparison or exploratory?
    - Main effects or interactions?
2. **What is my experimental design?**
    - How many factors?
    - Balanced or unbalanced?
    - Any batch effects?
3. **What are my reference groups?**
    - Control condition?
    - Wildtype genotype?
    - Order matters!

After Getting Results

4. **Do the results make biological sense?**
    - Known genes behaving as expected?
    - Reasonable number of DE genes?
    - Effect sizes plausible?
5. **What is the reproducibility?**
    - Can I reproduce this next week?
    - Did I document all parameters?
    - Can someone else run my code?
6. **What's the next step?**
    - Validate top genes?
    - Pathway analysis?
    - New experiment?

Summary

Take Home Messages

1. **Normalization is essential**
   - TMM handles library size + composition bias
   - Most methods assume most genes NOT DE
2. **Design matrices encode your experiment**
   - Each column = one parameter to test
   - Think about your reference groups!
3. **Linear models are flexible**
   - Can handle any design
   - Contrasts let you ask specific questions
4. **limma-voom combines best of both worlds**
   - Transforms counts for linear models
   - Borrows strength across genes
   - Weights handle heteroscedasticity

Take Home Messages (cont.)

5. **Think before clicking**
    - Plan your contrasts based on biology
    - Consider interactions carefully
    - Start simple, add complexity as needed
6. **Both statistics AND biology matter**
    - Significant ≠ Important
    - Effect size ≠ Certainty
    - Consider both logFC and p-value
7. **Reproducibility is critical**
    - Document everything
    - Share code, not just methods
    - Future you will thank you!

Ready for Hands-On Analysis!

What’s Next

In the practical session, you’ll:

1. ✓ Load and prepare RNA-seq count data
2. ✓ Apply TMM normalization
3. ✓ Create and interpret design matrices
4. ✓ Run limma-voom analysis
5. ✓ Compare edgeR vs limma-voom
6. ✓ Build and test contrast matrices
7. ✓ Interpret and visualize results
8. ✓ Export publication-ready figures

Now you’ll understand WHY each step matters!

Let’s start coding →

Additional Resources

limma User's Guide
https://bioconductor.org/packages/release/bioc/vignettes/limma/inst/doc/usersguide.pdf

edgeR User's Guide
https://bioconductor.org/packages/release/bioc/vignettes/edgeR/inst/doc/edgeRUsersGu

RNA-seq analysis is easy as 1-2-3 with limma, Glimma and edgeR
Law et al. 2016, *F1000Research*

From reads to genes to pathways
Law et al. 2016, *F1000Research*

Harvard Chan Bioinformatics Core Tutorials
https://hbctraining.github.io/

Our workshop materials
[Your GitHub repo link]